AI STUDY COMPANION USING RETRIEVAL-AUGMENTED GENERATION (RAG) AND GEMINI AI

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ABSTRACT

This project presents an AI-powered Study Companion developed using Retrieval-Augmented Generation (RAG), Google Gemini AI, FAISS, and Sentence Transformers to enhance students' learning experience. Traditional study methods often require students to manually search through lengthy notes and textbooks, making revision time-consuming and inefficient. The proposed system addresses this challenge by allowing users to upload study materials in PDF format and interact with them through natural language questions.

The uploaded PDF is processed by extracting text, cleaning unwanted characters, and dividing the content into smaller chunks. These chunks are converted into vector embeddings using the SentenceTransformer model and stored in a FAISS vector database for efficient semantic search. When a user asks a question, the application retrieves the most relevant information from the uploaded notes and provides it as context to Google Gemini 2.5 Flash, which generates accurate and context-aware responses. This Retrieval-Augmented Generation (RAG) approach significantly reduces hallucinations and improves answer reliability.

In addition to intelligent question answering, the application includes an AI Quiz Generator that automatically creates multiple-choice questions from the uploaded notes and a Flashcard Generator that produces concise question-and-answer pairs for quick revision. The system is implemented using Python and Streamlit, providing a simple, responsive, and user-friendly interface.

The AI Study Companion serves as an intelligent educational assistant that improves concept understanding, supports self-learning, and simplifies exam preparation. By integrating modern AI technologies such as Natural Language Processing, vector databases, semantic search, and Large Language Models, the project demonstrates a practical application of Artificial Intelligence in education and provides students with a personalized and efficient digital learning platform.

1. INTRODUCTION

Artificial Intelligence has transformed many domains, including education, by enabling intelligent systems that assist users in learning and decision-making. Traditional methods of studying from printed books and handwritten notes require significant time to locate relevant information, especially during examinations. Large Language Models (LLMs) have made conversational learning possible; however, they may generate incorrect or outdated information if not provided with reliable context.

The objective of this project is to develop an AI-powered Study Companion capable of answering questions directly from students' uploaded study notes. The application combines Retrieval-Augmented Generation (RAG), semantic search, and Google's Gemini AI to generate accurate, context-aware responses. Instead of relying solely on the language model's internal knowledge, the system retrieves relevant content from uploaded PDFs before generating answers.

The project also provides additional learning tools such as automatic quiz generation and flashcard creation, helping students revise concepts more effectively. Developed using Python, Streamlit, FAISS, Sentence Transformers, and Google Gemini AI, the system demonstrates how modern AI technologies can improve digital learning and personalized education.

2. PROBLEM STATEMENT

Students often spend considerable time searching through textbooks, lecture notes, and PDFs to find specific information during study and revision. Traditional search methods are inefficient because they rely on keyword matching rather than understanding the meaning of the content. General AI chatbots may also provide incorrect or unrelated answers when they are not given the appropriate study material.

The objective of this project is to develop an intelligent study assistant capable of understanding uploaded PDF notes, retrieving relevant information using semantic search, and generating accurate answers through Retrieval-Augmented Generation (RAG). The system also aims to improve revision by automatically generating quizzes and flashcards from the uploaded study material.

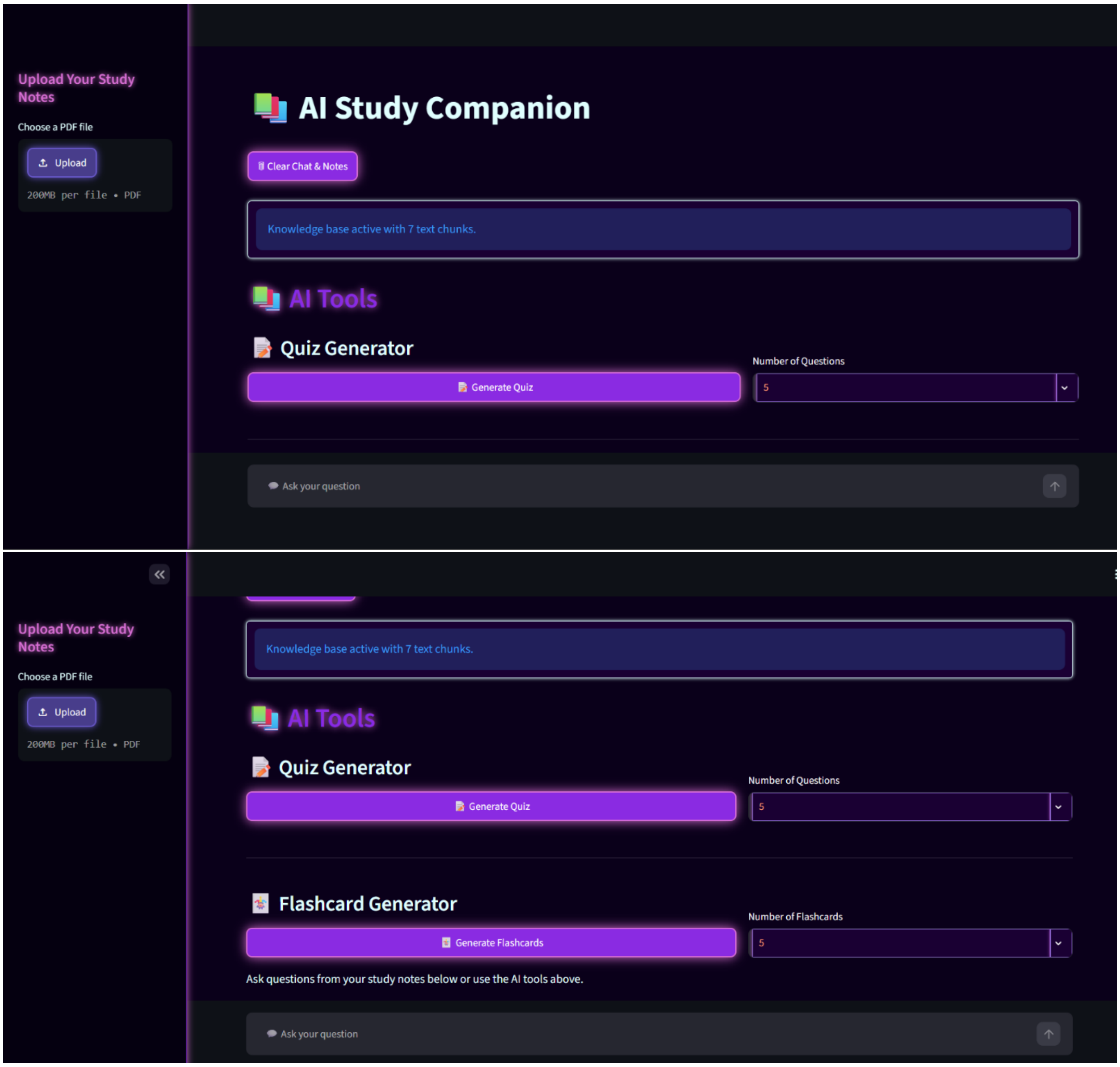
3. METHODOLOGY

- **PDF Upload using Streamlit**
- **Text Extraction using PyPDF**
- **Text Cleaning and Preprocessing**
- **Text Chunking**
- **Embedding Generation using SentenceTransformer**
- **Vector Storage using FAISS**
- **Semantic Search**
- **Retrieval-Augmented Generation (RAG)**
- **Response Generation using Google Gemini 2.5 Flash**

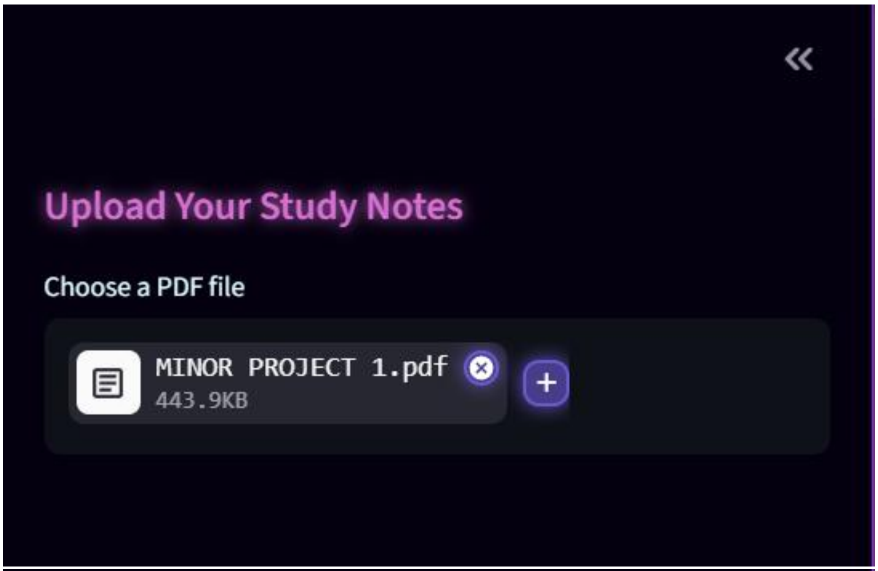
- Quiz Generation
- Flashcard Generation

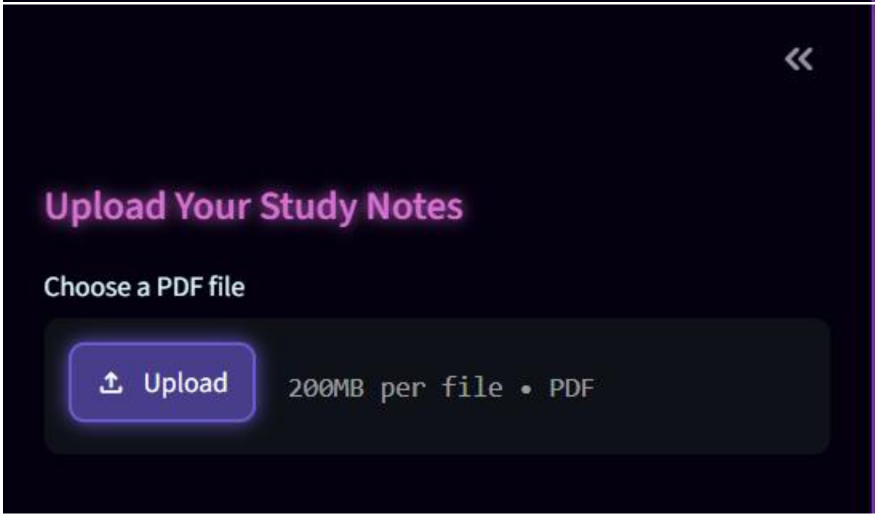
4. RESULTS AND DISCUSSION

1: Home Page of AI Study Companion

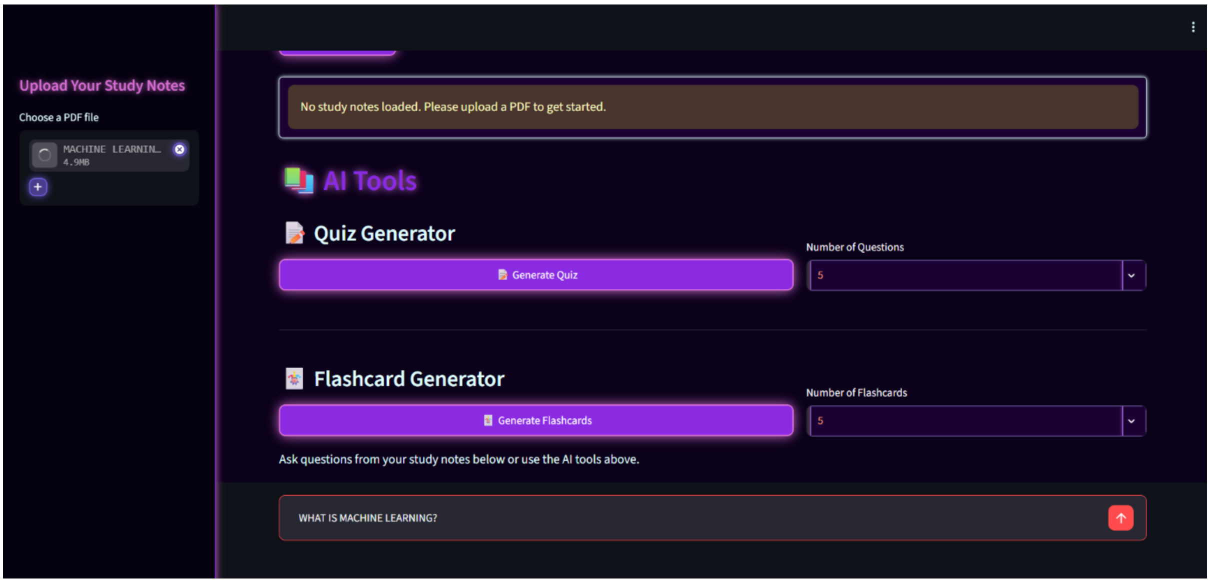


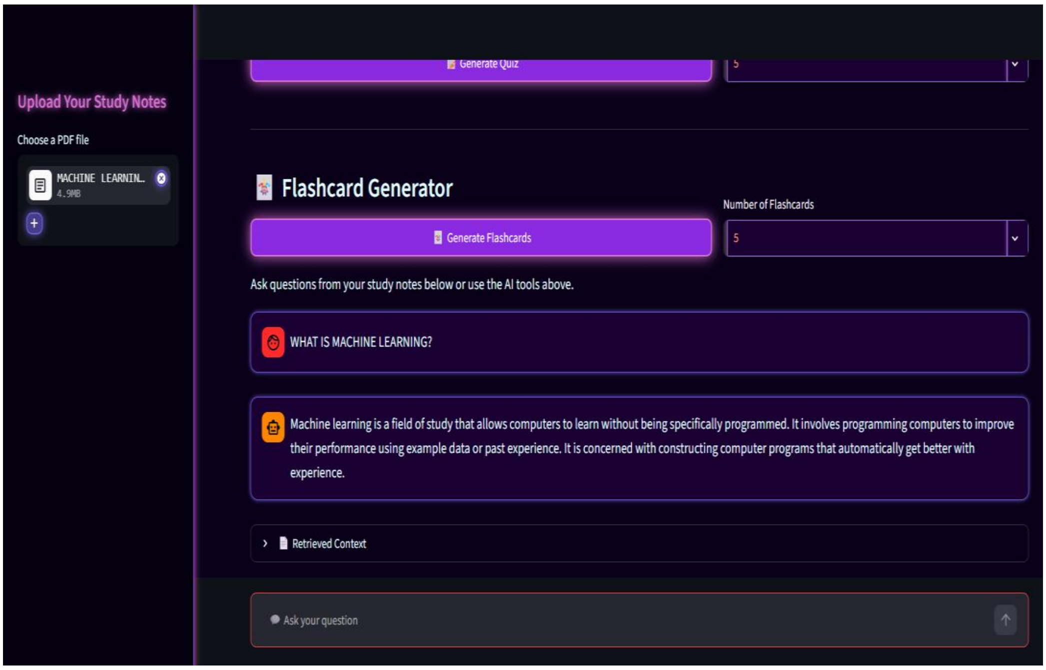
2: PDF Upload Interface



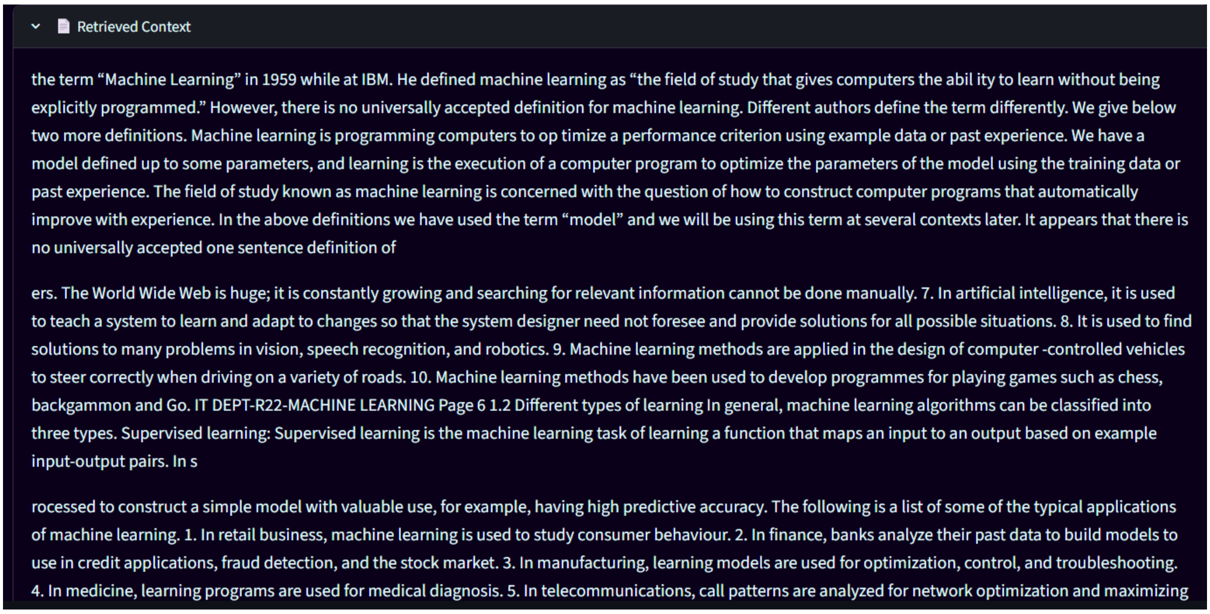


3: AI Question Answering





4: Retrieved Context Display

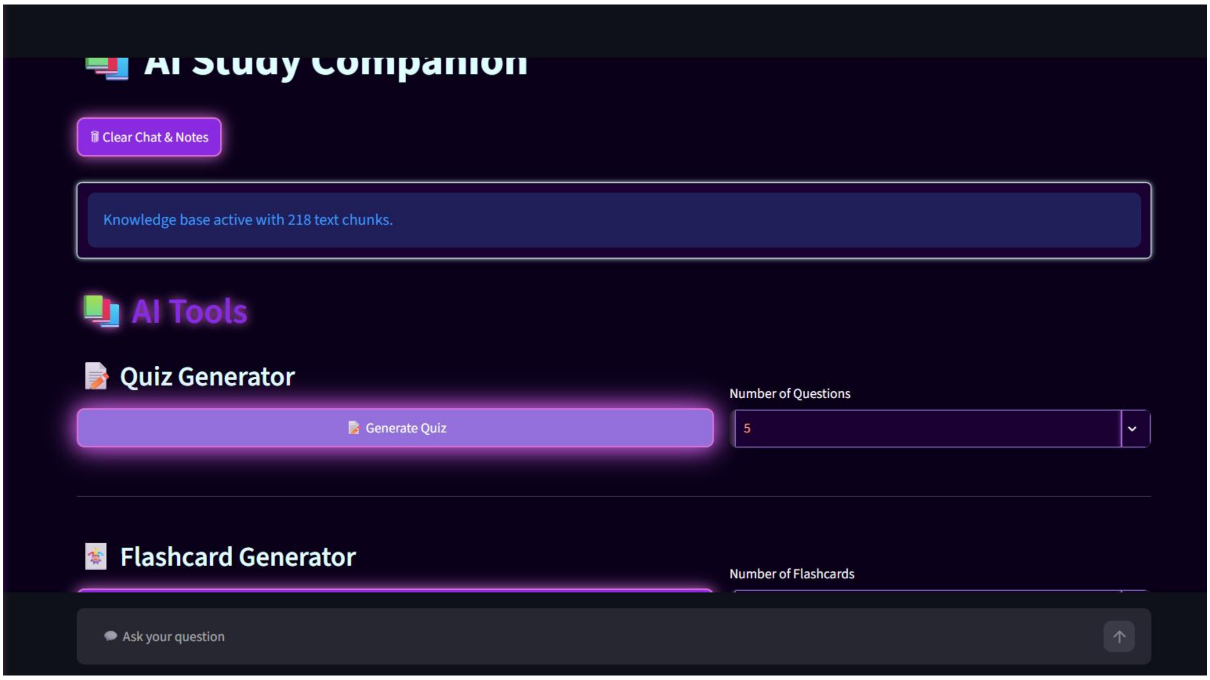


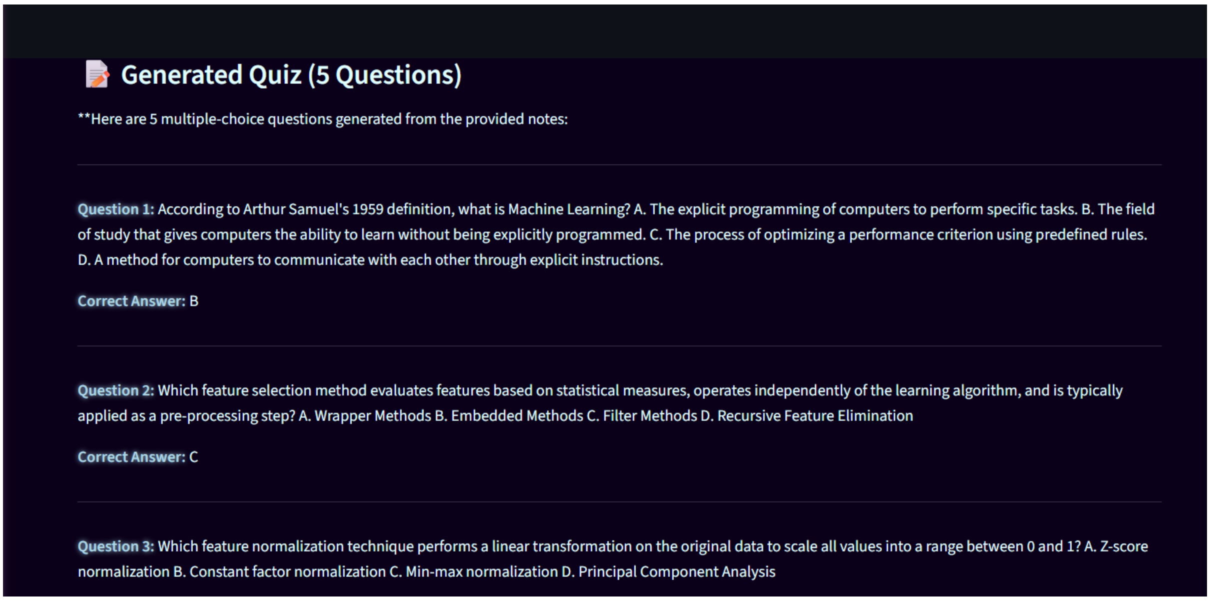
use in credit applications, fraud detection, and the stock market. 3. In manufacturing, learning models are used for optimization, control, and troubleshooting. 4. In medicine, learning programs are used for medical diagnosis. 5. In telecommunications, call patterns are analyzed for network optimization and maximizing the quality of service. 6. In science, large amounts of data in physics, astronomy, and biology can only be analyzed fast enough by computers. The World Wide Web is huge; it is constantly growing and searching for relevant information cannot be done manually. 7. In artificial intelligence, it is used to teach a system to learn and ad

f turning the knowledge about stored data into a form that can be utilized for future action. These actions are to be c arried out on tasks that are similar, but not identical, to those what have been seen before. In generalization, the goal is to discover those properties of the data that will be most relevant to future tasks. Evaluation IT DEPT-R22-MACHINE LEARNING Page 5 Evaluation is the last component o f the learning process. It is the process of giving feedback to the user to measure the utility of the learned knowledge. This feedback is then utilised to effect improvements in the whole learning process. 1.1.3 Applications of machine learning Application of machine learning methods to large databases is called data mining. In data mining, a large volume of data is processed to construct a simple model with valuable use, for example, having high predictive accuracy. The following is a list of some of the typical applications of machine learning. 1. In retail

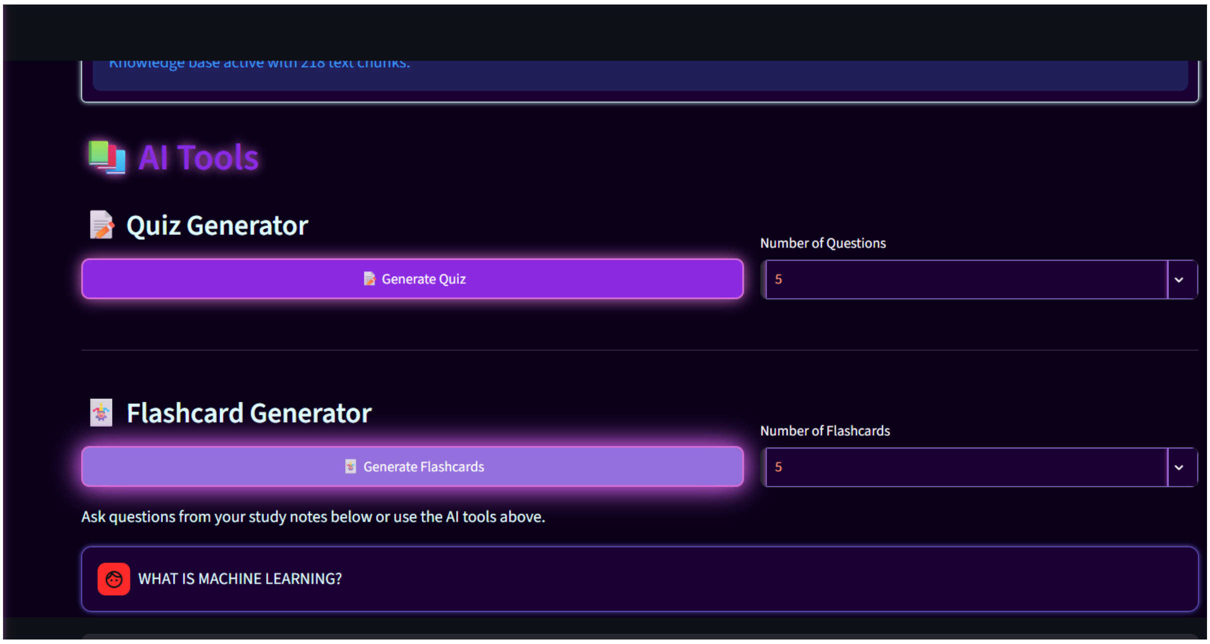
oss validation 83 18 Bias-Variance Trade off 89 19 Regularization 92 20 Overfitting, Underfitting 96 21 UNIT-V Clustering 102 22 Markov Decision Process 124 23 Q-learning 127 IT DEPT-R22-MACHINE LEARNING Page 2 UNIT-I Introduction: Introduction to Machine learning, Supervised learning, Unsupervised learning, Reinforcement learning. Deep learning. Feature Selection: Filter, Wrapper Embedded methods. Feature Normalization:- min-max normalization, z-score normalization, and constant factor normalization Introduction to Dimensionality Reduction : Principal Component Analysis(PCA), Linear Discriminant Analysis(LDA) Introduction 1.1 Definition of Machine Learning Arthur Samuel, an early American leader in the field of computer gaming and artificial intelligence, coined the term “Machine Learning” in 1959 while at IBM. He defined machine learning as “the field of study that gives computers the abil ity to learn without being explicitly programmed.” However, there

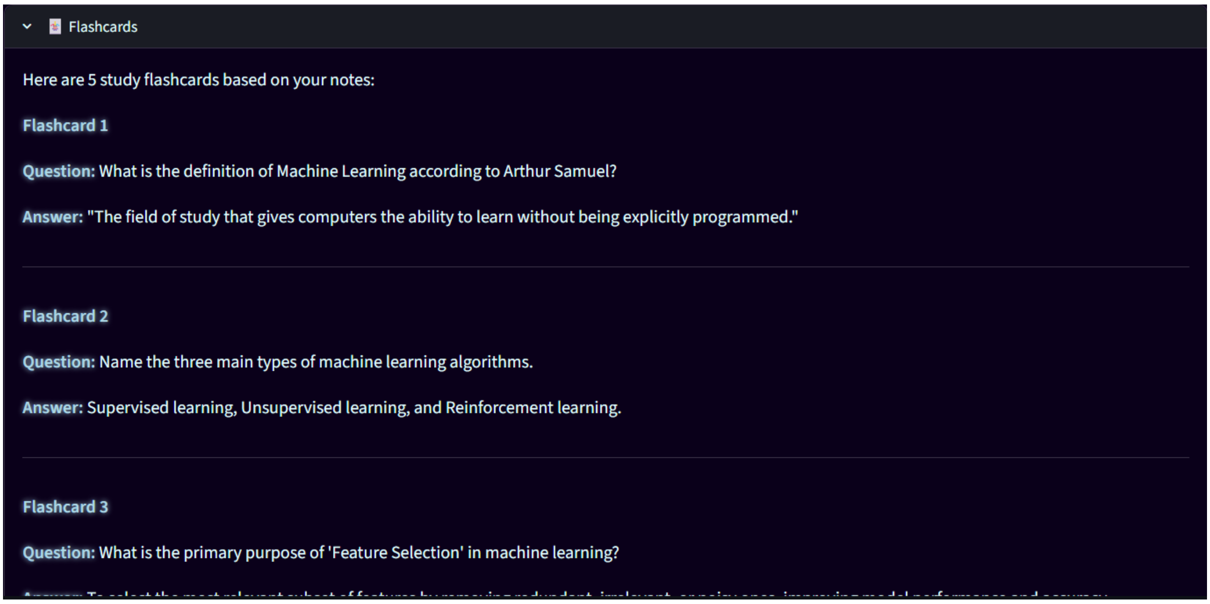
5: Quiz Generator





6: Flashcard Generator





The developed AI Study Companion successfully answered questions based on uploaded study notes by combining semantic retrieval with Google's Gemini AI model. The FAISS vector database enabled fast retrieval of relevant information, while Sentence Transformers generated meaningful text embeddings that improved semantic search accuracy. The Retrieval-Augmented Generation (RAG) architecture reduced irrelevant responses by ensuring that answers were generated using the uploaded study material rather than relying solely on the language model.

The Quiz Generator and Flashcard Generator further enhanced the learning experience by providing automatic revision material from the uploaded notes. The Streamlit-based interface offered a responsive and user-friendly environment, making the application suitable for students with minimal technical knowledge. Overall, the system demonstrated accurate context retrieval, efficient document processing, and interactive AI-assisted learning.

5. CONCLUSION

The AI Study Companion successfully demonstrates the integration of Retrieval-Augmented Generation, semantic search, and Large Language Models to create an intelligent educational assistant. By allowing users to upload PDF study materials and ask questions in natural language, the application provides accurate and context-aware responses while minimizing irrelevant information.

Additional features such as Quiz Generation and Flashcard Generation make the system useful for exam preparation and concept revision. The project highlights the practical application of Artificial Intelligence, Natural Language Processing, vector databases, and modern LLMs in the education domain. Future enhancements may include multilingual support, voice interaction, cloud deployment, OCR support for scanned PDFs, and user authentication.

6. REFERENCES

- Google Gemini API Documentation
- Streamlit Documentation
- FAISS Documentation (Facebook AI Similarity Search)
- Sentence Transformers Documentation
- PyPDF Documentation
- Python Official Documentation
- Hugging Face Sentence Transformers
- Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS.